DAVID MENDOZA

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EDUCATION

University of Houston, Cullen College of Engineering — Houston, Texas
Bachelor of Computer Engineering Technology | Dean's List **December 2026**

EXPERIENCE

TOC/CIR Assistant | FIFA World Cup 2026 **June 2026 – July 2026**
FIFA - hbs, Houston, Texas

- Facilitated the integration of Media Partners' (MPs) broadcasting equipment with venue infrastructure, routing complex fiber optic looms through the Center Interconnection Room (CIR).
- Monitored live feed stability and signal distribution across global networks, acting as a critical technical checkpoint between the Technical Operations Centre (TOC) and Technical Interconnection Room (TIR).
- Ensured uninterrupted broadcast operations by verifying that all media partners were successfully sending and receiving high-quality feeds.
- Gained practical exposure to live broadcast engineering systems, thanks to NEP and their Outside Broadcast (OB) truck; and signal processing, and transmission to geo-locked satellites thanks to ARCTEK.

Paint Associate **July 2020 - Present**
The Home Depot, Houston, Texas

- Managed inventory logistics and translated complex technical product specifications for B2B and retail clients, ensuring project requirements were met and improving operational efficiency.

PROJECTS

AI Robotic Arm

- Developed an AI-driven robotic arm using an NVIDIA Jetson Orin Nano, leveraging LeRobot and CUDA integration to accelerate machine learning model training for automated robotics tasks.
- Interfaced sensor arrays with L298N motor controllers, implementing PID control algorithms to ensure precise, real-time physical adjustments and smooth joint movements.

Reaction Timer

- Engineered a digital reaction timer on an FPGA, authoring modular VHDL/Verilog code for a finite state machine (FSM) to measure and display human response times with millisecond precision.

Tiny-VGG

- Developed a CNN based on the VGG architecture using PyTorch and CUDA for accelerated training on an NVIDIA GPU, achieving 80% accuracy on image classification tasks.

SKILLS

- **Programming & Web:** Python, Verilog, C, C++, HTML, CSS, JavaScript
- **Software & Tools:** MATLAB, Multisim, LTspice, LabVIEW, Fusion 360, Quartus, Git.
- **Engineering Skills:** Circuit Design, System Integration, FPGA, PID Algorithms, Deep Learning.
- **Languages:** Bilingual (English/Spanish).